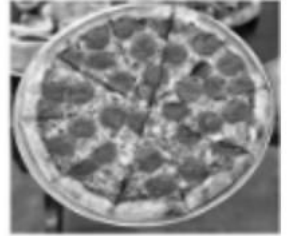


Geometry
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Use the given context to answer questions 1-10.

Eduardo orders a small circular pizza that has a 10-inch (in.) diameter and is cut into 6 triangular slices.



1. What is the approximate area of a 10-inch pizza? Use 3.14 for π .
2. If the slices are cut precisely and equally, what fraction and percentage of the pizza is 3 slices?
3. What is the area of 3 slices?
4. What is the central angle of 3 slices?
5. If the slices are cut precisely and equally, what fraction of the pizza is 5 slices?

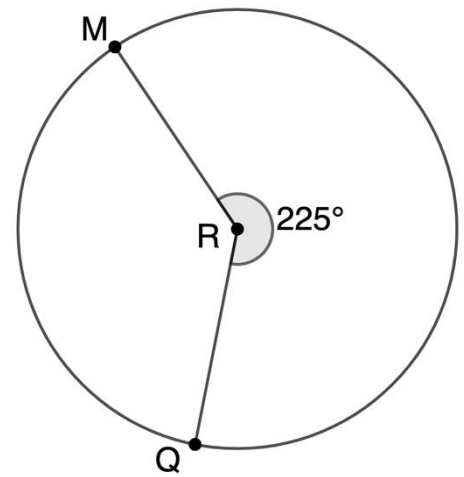
6. What is the area of 5 slices?
7. What is the central angle of 5 slices?
8. If the slices are cut precisely and equally, what fraction of the pizza is 1 slice?
9. What is the area of 1 slice?
10. What is the central angle of 1 slice?

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Lesson 2 Practice

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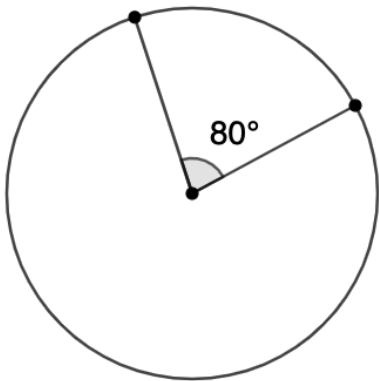
For questions 1-2, Circle R is shown where points M and Q are on the circle. The measure of central angle QRM is 225° .

1. What is the ratio of the central angle, $\angle QRM$, to the whole circle? Simplify the ratio.
2. What fraction of the circle's circumference does $\widehat{MQ}$ represent?

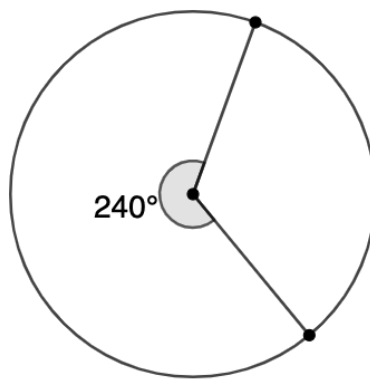


For questions 3-8, find the ratio of the arc length to the circumference of the circle.

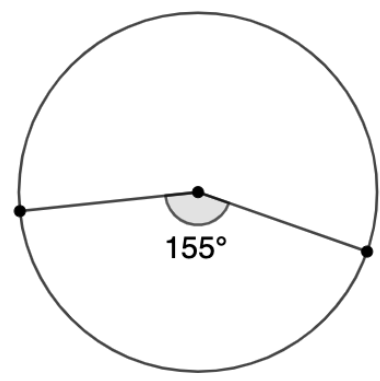
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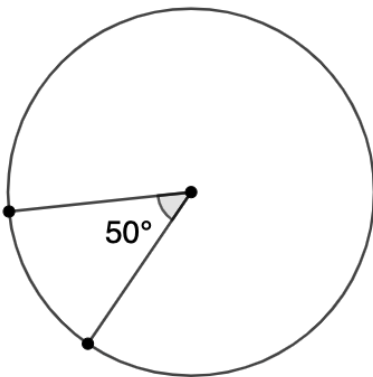
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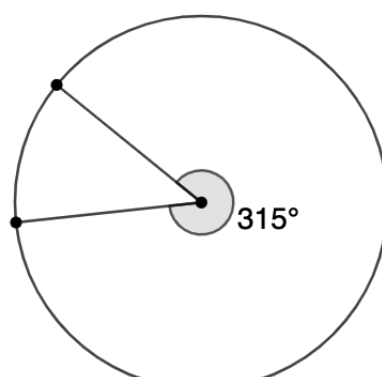
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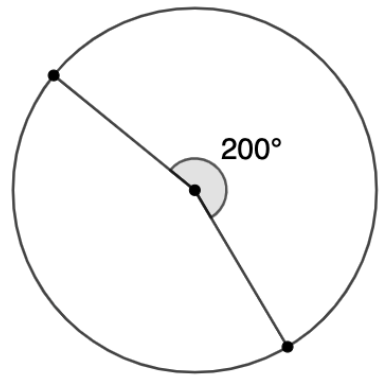
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7.

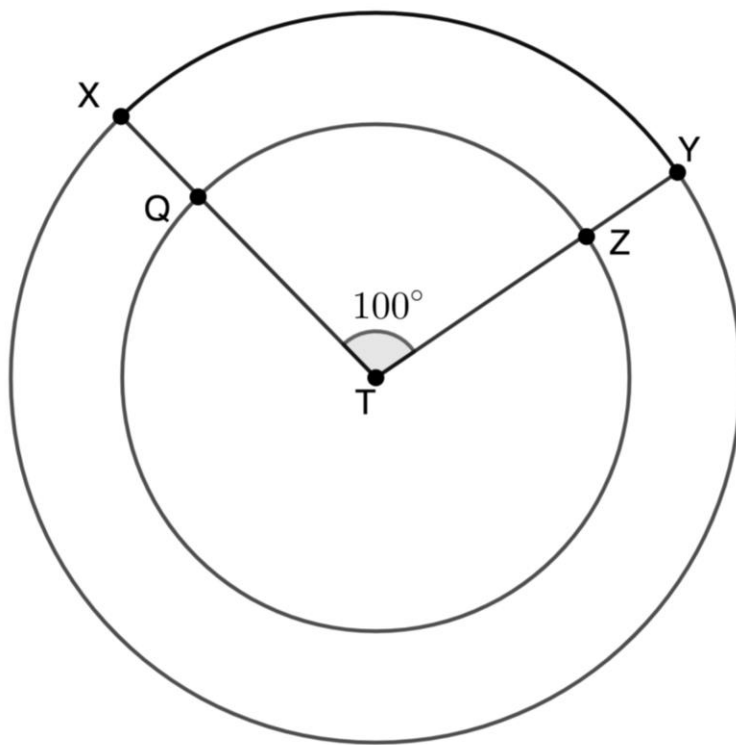


8.



For questions 9-10, consider the following information.

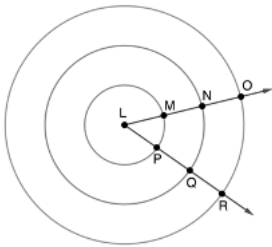
Points X and Y lie on Circle T . The radius $\overline{TX}$ measures 10 inches (in.). A smaller Circle T with a radius measure of 7 in. rests inside the larger Circle T , where points Q and Z lie on the smaller circle. The smaller Circle T has central angle $\angle QTZ$ that measures 100° and the larger Circle T has central angle $\angle XTY$ that measures 100° .



9. What is the ratio of the arc length to circumference of Circle T with radius measure of 10 in.?

10. What is the ratio of the arc length to circumference of Circle T with radius measure of 7 in.?

For questions 1-6, concentric circles with center L are shown. Points M and P lay on the innermost Circle L , and has radius $LM = 3$ units. Points N and Q lay on the middle Circle L , and has radius $LN = 6$ units. Points O and R lay on the outer Circle L , and has radius $LO = 9$ units.



Determine the circumference for each circle. Leave your answer in terms of π .

	Circle	Circumference
1.	Innermost Circle	
2.	Middle Circle	
3.	Outer Circle	

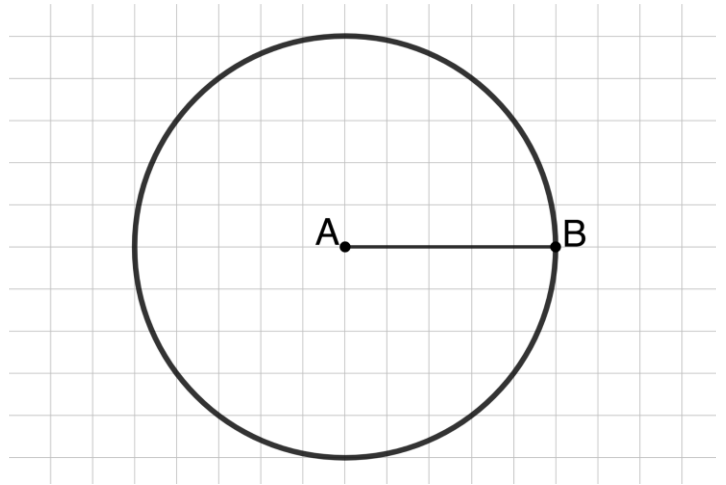
Complete the table.

	Circle	Ratio of Circumference to Radius
4.	Innermost Circle	
5.	Middle Circle	
6.	Outer Circle	

7. What would be the ratio of the circumference to the radius for a circle whose center is W and whose radius has a measurement of 16?

Circle A is shown where $\overline{AB}$ is a radius.

8. Label point C on the circumference of circle A so that the resulting arc has an approximate length of 1 radius.



9. Describe how you determined the location of point C .

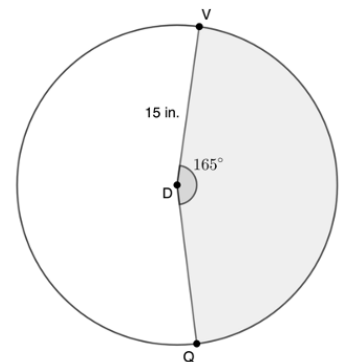
10. Label point W on the circumference of circle A so that the resulting arc has an approximate length of 2 radii.

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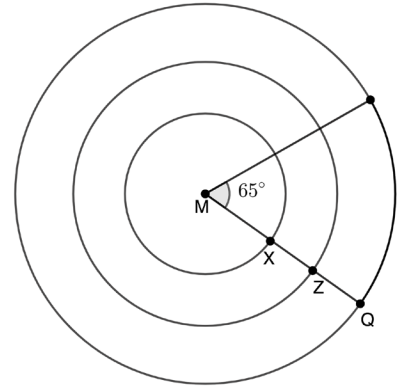
1. Circle C has a diameter of 18 inches (in.). Points T and S are on the circle and form the sector TCS . The central angle $\angle TCS$ in Circle C measures 120° . Determine the area of the sector. Express your answer in terms of π .
2. The area of a sector with a radius of 5 centimeters (cm) is 15.7 square cm. Calculate the approximate measure of the central angle of the sector in degrees.

3. Circle D has radius $\overline{DV}$ that measures 15 inches (in.) with central angle $\angle VDQ$, where $m\angle VDQ = 165^\circ$. Find the area of the shaded sector. Round your answer to the nearest hundredth.



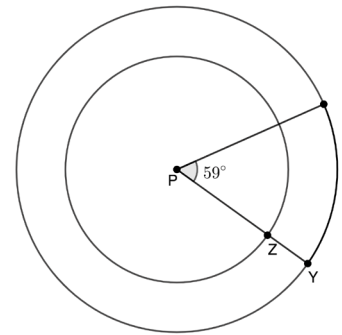
4. Circle K has a radius of 7 inches (in.). Points R and Y are on the circle and form the sector RKY . The central angle $\angle RKY$ in Circle K measures 145° . Determine the area of the sector. Express your answer in terms of π .
5. The area of a sector with a diameter of 22 feet (ft) is 27.44 square ft. Calculate the approximate measure of the central angle of the sector in degrees.

For questions 6-8, concentric circles with center M are shown with $m\angle M = 65^\circ$. The smallest Circle M has a radius $\overline{MX}$ that measures 10 yards (yd). The largest Circle M has radius $\overline{MQ}$ and the middle Circle M has radius $\overline{MZ}$. Each circle has a radius that is 10 yd longer than the immediate prior circle.



6. What is the arc length of the circle with radius $\overline{MX}$?
7. What is the arc length of the circle with radius $\overline{MZ}$?
8. What is the arc length of the circle with radius $\overline{MQ}$?

For questions 9-10, concentric circles with center P are shown with $m\angle P = 59^\circ$. The smallest Circle P has a radius $\overline{PZ}$ that measures 9 meters (m) and the largest Circle P has radius $\overline{PY}$. The radius of the largest Circle P has a radius 4 m larger than the smaller Circle P .



9. What is the arc length of the circle with radius $\overline{PZ}$?
10. What is the arc length of the circle with radius $\overline{PY}$?

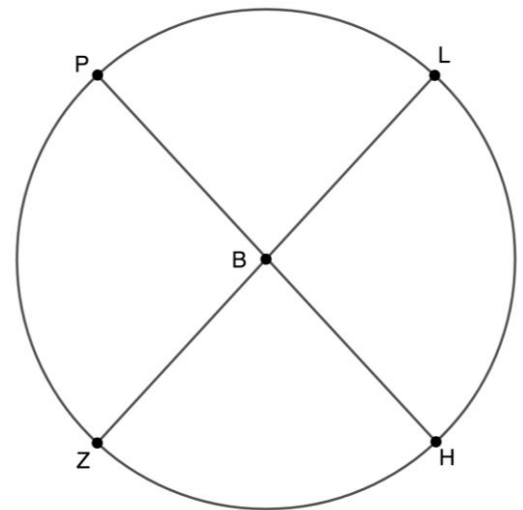
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Lesson 5 Practice

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- Circle C has points X, N , and P on the circle with $m\angle NCX = 47.1^\circ$. If $m\widehat{NP} = 116.9^\circ$ and $\widehat{NP}$ does not intersect $\widehat{NX}$, what is $m\angle PCX$?
- Circle S has radii $\overline{ST}$ and $\overline{SR}$. Find the value of x if $m\angle RST = 135^\circ$ and $m\widehat{RT} = (6x + 9)^\circ$.

For questions 3-4, Circle B has diameters $\overline{PH}$ and $\overline{LZ}$ that both intersect at point B .

- Given $m\widehat{HZ} = 72^\circ$, find the $m\widehat{PL}$.
- Given $m\widehat{HL} = 108^\circ$, find the $m\widehat{ZP}$.

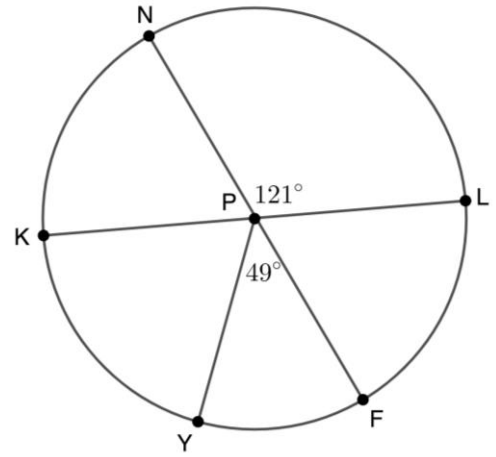


For questions 5-6, consider the following context.

Lorenzo is creating a folding fan. He needs to determine the measure of the arc of the fan before he can cut the paper. He uses the expression $(6x + 110)^\circ$ as the measure of the central angle, $\angle ARQ$. For the $m\widehat{QA}$, he uses the expression $(24x + 38)^\circ$.

- What is the value of x ?
- What is the degree measure of the arc of the fan, $\widehat{QA}$?

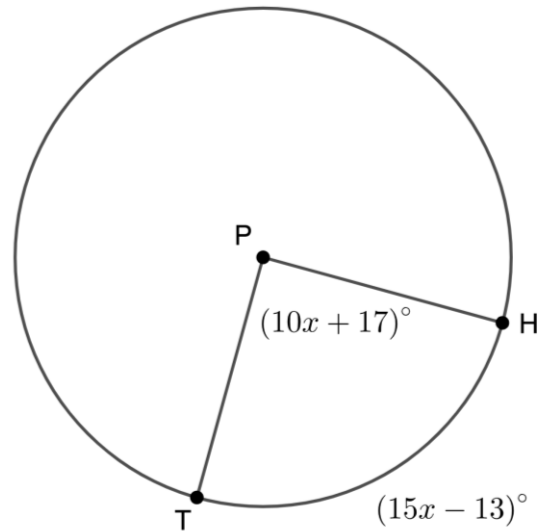
7. In Circle P , $m\angle NPL = 121^\circ$ and $m\angle FPY = 49^\circ$. Calculate the $m\widehat{NLY}$, given that P is the center and $\overline{NF}$ and $\overline{KL}$ are diameters.



8. Circle S has points K, D , and Y on the circle.
 $m\angle YSD = 71^\circ$, $m\widehat{KY} = 132^\circ$, and $\widehat{KY}$ does not intersect $\widehat{YD}$.
 Find $m\angle KSD$.

For questions 9-10, consider the following context.

A horse is tied to a post at point P and is grazing in a circular pattern. She decides to walk from her position at point H to the trough of water at point T along the circumference of the circular area she can roam. The measure of the central angle, $\angle TPH$, is $(10x + 17)^\circ$, and the measure of the arc that the horse walked, $\widehat{TH}$, is $(15x - 13)^\circ$.



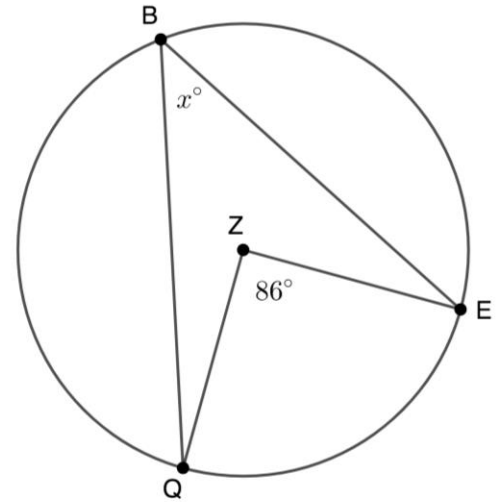
9. Find the value of x .

10. Find the measure of the arc that the horse walked, $\widehat{TH}$.

Geometry
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Lesson 6 Practice

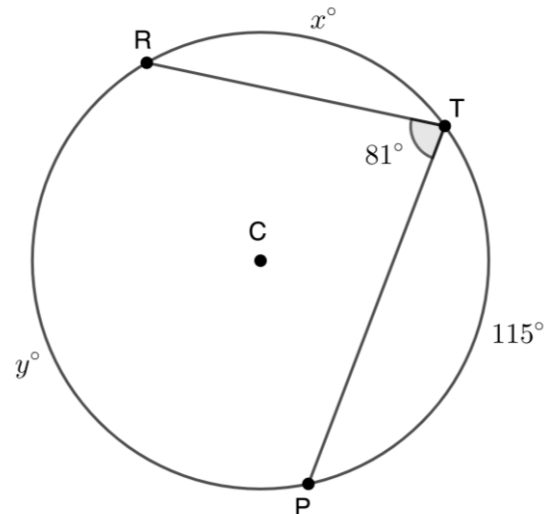
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1. Circle Z is shown where $\angle EZQ$ is a central angle, $\angle EBQ$ is inscribed in the circle, and $m\angle EZQ = 86^\circ$. Find $m\angle EBQ$.



For questions 2-3, Circle C has points R , T , and P on the circle. $\angle RTP$ is inscribed in Circle C measuring 81° , and the $m\widehat{TP} = 115^\circ$.

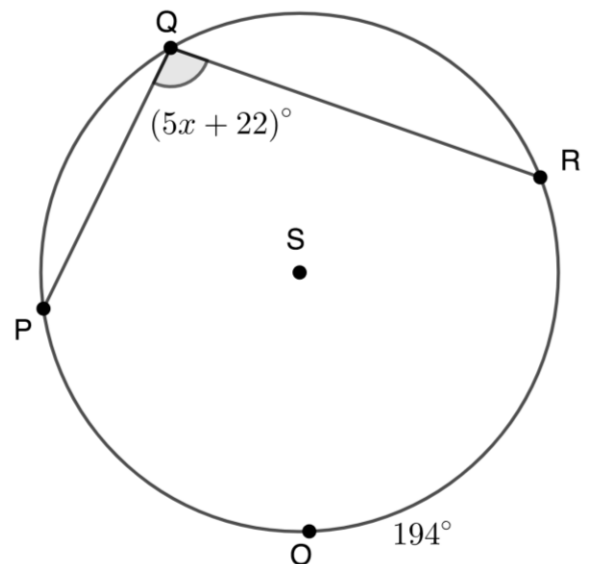
2. Find the $m\widehat{RP}$.



3. Find the $m\widehat{RT}$.

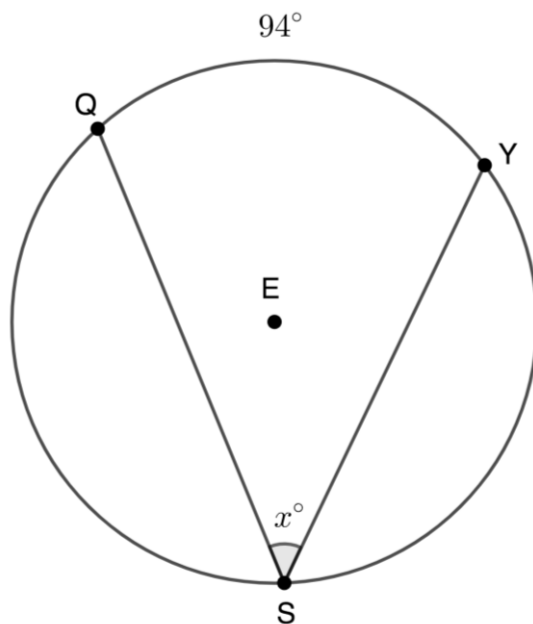
For questions 4-5, points O , P , Q , and R lie on Circle S . $\angle PQR$ is inscribed in the circle, the $m\angle PQR = (5x + 22)^\circ$, and the $m\widehat{ROP} = 194^\circ$.

4. Find the $m\angle PQR$.



5. Find the value of x .

6. Circle E is shown, where points Q , Y , and S are on the circle. $\angle QSY$ is inscribed in Circle E with $m\widehat{QY} = 94^\circ$. Find the value of x .



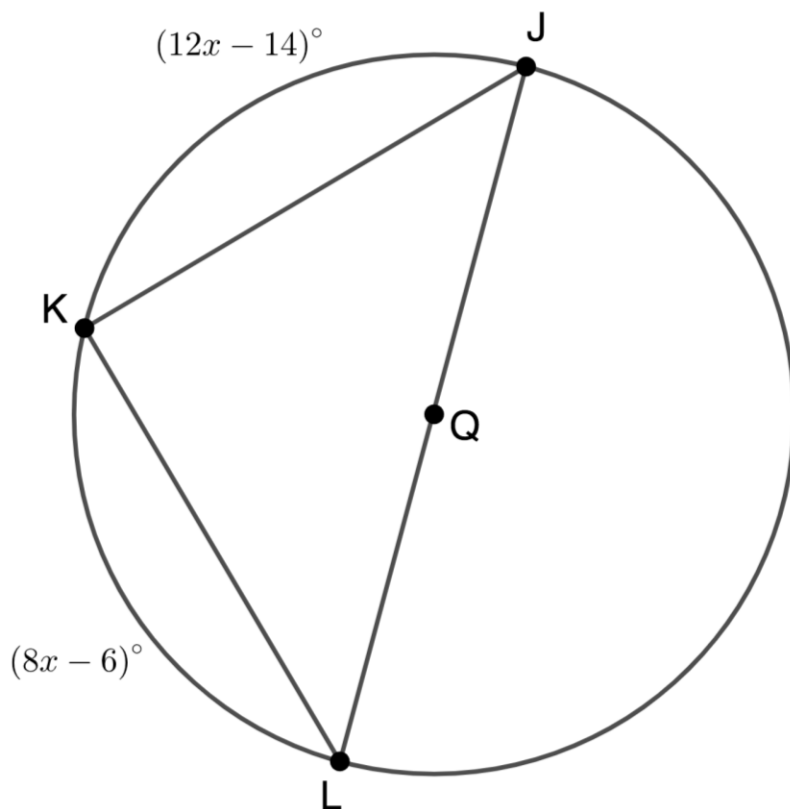
For questions 7-10, Circle Q is shown with inscribed angle $\angle LKJ$ and diameter $\overline{JL}$. The $m\widehat{KJ} = (12x - 14)^\circ$ and the $m\widehat{KL} = (8x - 6)^\circ$.

7. Find the value of x .

8. Find the $m\angle KJL$.

9. Find the $m\angle KLJ$.

10. Find the $m\widehat{KJ}$.

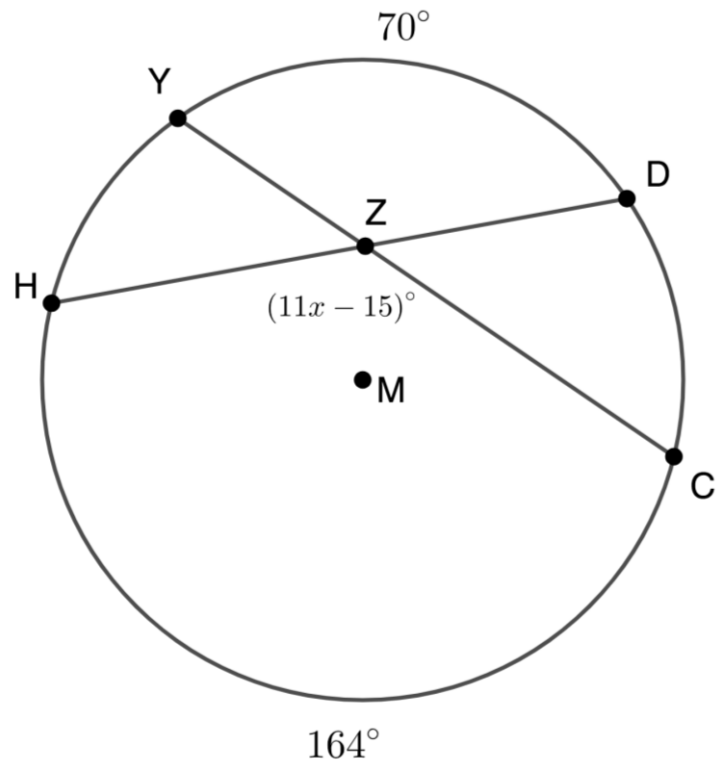


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Lesson 7 Practice

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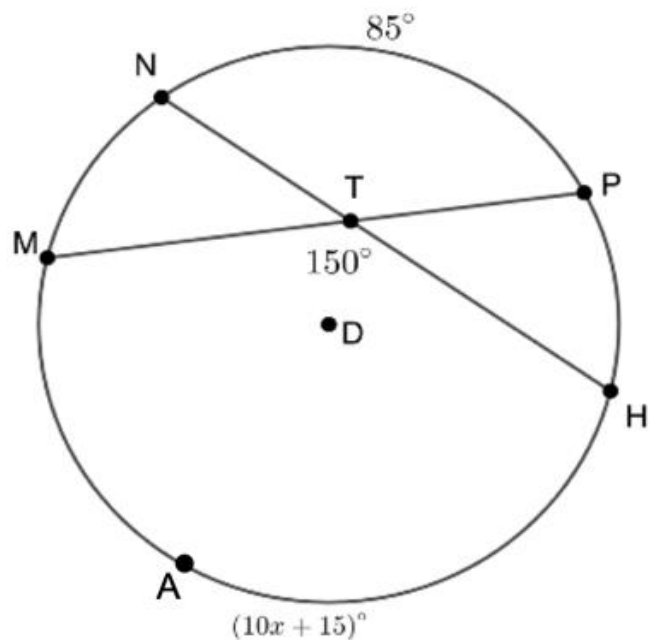
For questions 1-3, Circle M is shown, where chords $\overline{YC}$ and $\overline{HD}$ intersect at point Z , $m\widehat{CH} = 164^\circ$, $m\widehat{YD} = 70^\circ$, and $m\angle HZC = (11x - 15)^\circ$.

- Find the $m\angle HZC$.
- Write an equation to find the value of x .
- Find the value of x .

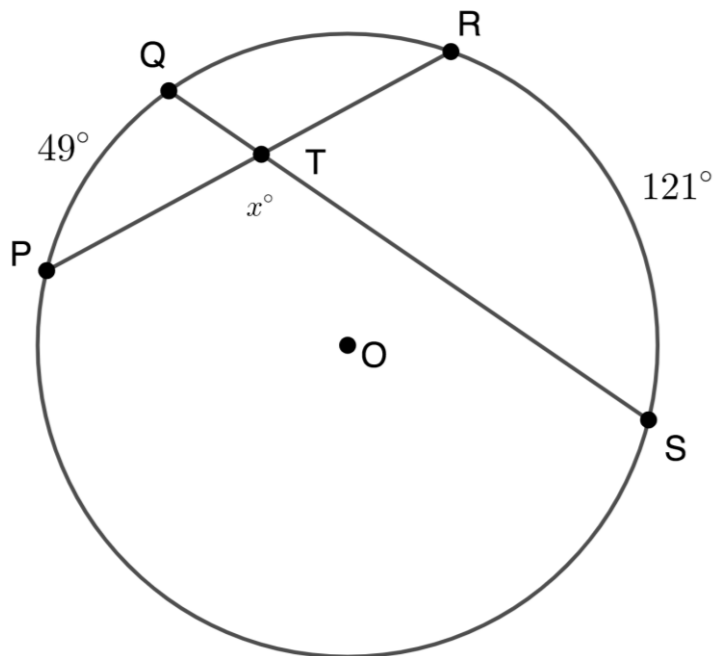


For questions 4-6, Circle D is shown, where chords $\overline{NH}$ and $\overline{MP}$ intersect at point T , $m\widehat{NP} = 85^\circ$, $m\widehat{MH} = (10x + 15)^\circ$, and $m\angle MTH = 150^\circ$.

- Write an equation to find the value of x .
- Find the value of x .
- Find the $m\widehat{MH}$.

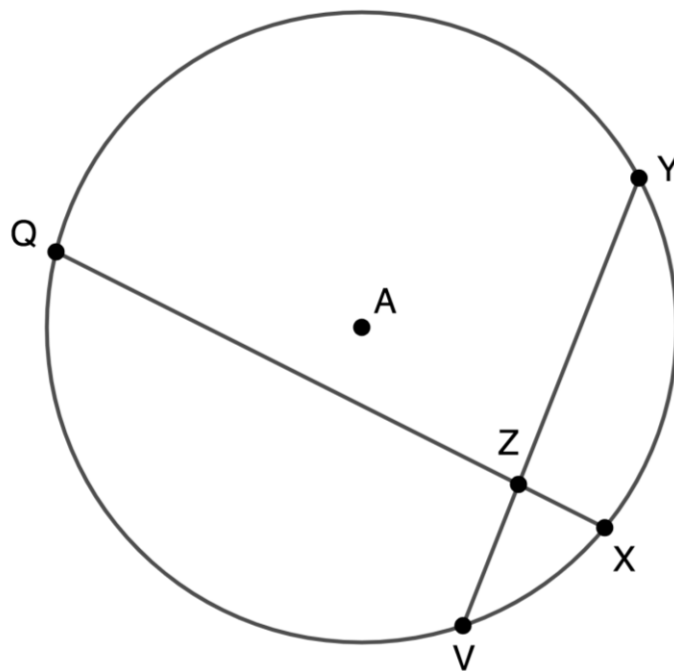


7. Circle O is shown, where chords $\overline{QS}$ and $\overline{RP}$ intersect at point T . Find the value of x for $\angle PTS$, given $m\widehat{RS} = 121^\circ$ and $m\widehat{PQ} = 49^\circ$.



For questions 8-10, Circle A is shown, where chords $\overline{QX}$ and $\overline{YV}$ intersect at point Z .

8. If $m\widehat{QY} = 137^\circ$ and $m\widehat{VX} = 35^\circ$, what is the measure of $\angle QZY$?



9. If $m\widehat{QV} = 141^\circ$ and $m\widehat{YX} = 47^\circ$, what is the measure of $\angle QZY$?

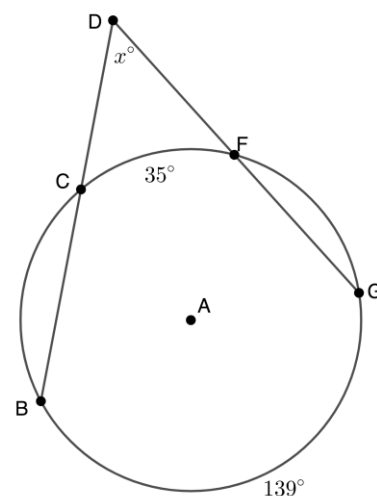
10. If $m\widehat{VX} = 41^\circ$ and $m\angle QZY = 84^\circ$, what is the measure of $\widehat{QY}$?

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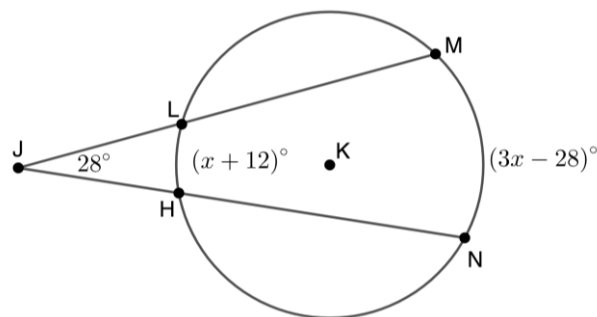
For questions 1-2, Circle A has points B , C , F , and G on the circle. Secant lines $\overline{BD}$ and $\overline{GD}$ intersect at point D outside the circle. The $m\widehat{BG} = 139^\circ$, $m\widehat{CF} = 35^\circ$, and $m\angle CDF = x^\circ$.

- Write a formula that can be used to find the value of x .



- Find the value of x .

For questions 3-5, Circle K has points N , H , L , and M on the circle. Secant lines $\overline{NJ}$ and $\overline{MJ}$ intersect at point J outside the circle. The $m\widehat{LH} = (x + 12)^\circ$, $m\widehat{MN} = (3x - 28)^\circ$, and $m\angle HJL = 28^\circ$.



Not drawn to scale

- Find the $m\widehat{MN}$.

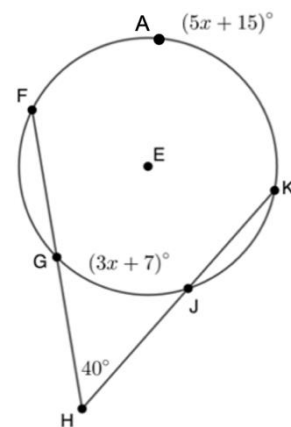
- Find the $m\widehat{LH}$.

For questions 6-8, Circle E has points F , G , J , and K on the circle. Secant lines $\overline{FH}$ and $\overline{KH}$ intersect at point H outside the circle. The $m\widehat{FAK} = (5x + 15)^\circ$, $m\widehat{GJ} = (3x + 7)^\circ$, and $m\angle GHJ = 40^\circ$.

6. Find the value of x .

7. Find the $m\widehat{GJ}$.

8. Find the $m\widehat{FAK}$.

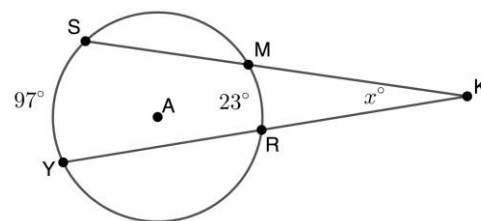


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For questions 9-10, Circle A has points S , M , Y , and R on the circle. Secant lines $\overline{SK}$ and $\overline{YK}$ intersect at point K outside the circle. The $m\widehat{SY} = 97^\circ$, $m\widehat{MR} = 23^\circ$, and $m\angle MKR = x^\circ$.

9. Write a formula that can be used to find the value of x .

10. Find the value of x .

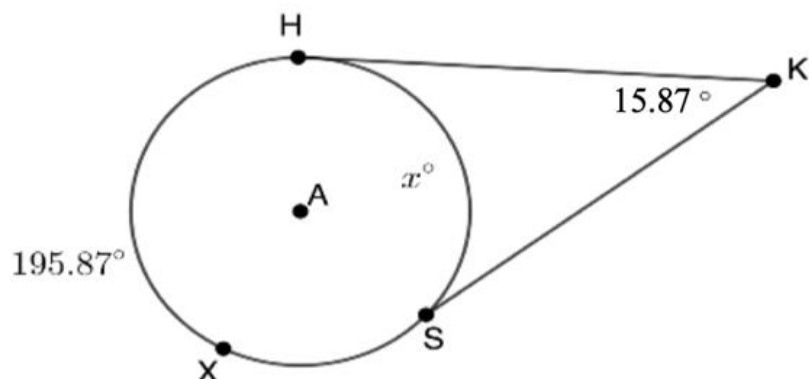


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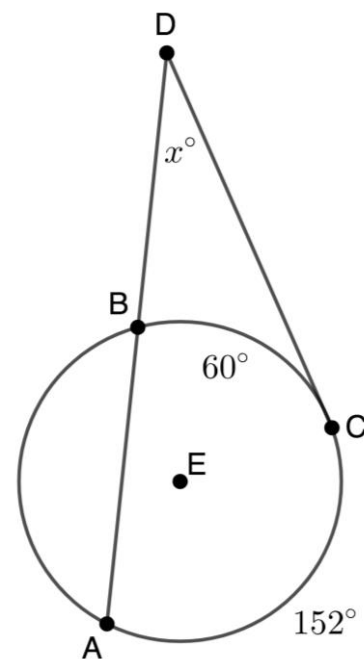
For questions 1-2, Circle A has points H, X , and S on the circle. Tangents $\overline{HK}$ and $\overline{SK}$ intersect at point K outside the circle. The $m\widehat{HXS} = 195.87^\circ$, $m\widehat{HS} = x^\circ$, and $m\angle HKS = 15.87^\circ$.



- Write an equation that can be used to find the value of x .

- Find the value of x .

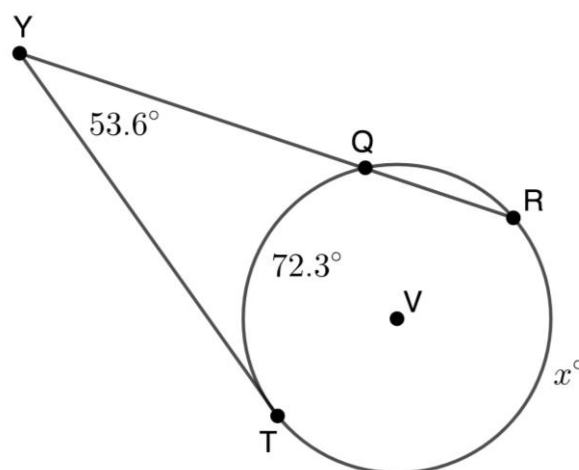
For questions 3-4, Circle E has points A, B , and C on the circle. Tangent $\overline{CD}$ and secant $\overline{AD}$ intersect at point D outside the circle. The $m\widehat{AC} = 152^\circ$, $m\widehat{BC} = 60^\circ$, and $m\angle BDC = x^\circ$.



- Write an equation that can be used to find the value of x .

- Find the value of x .

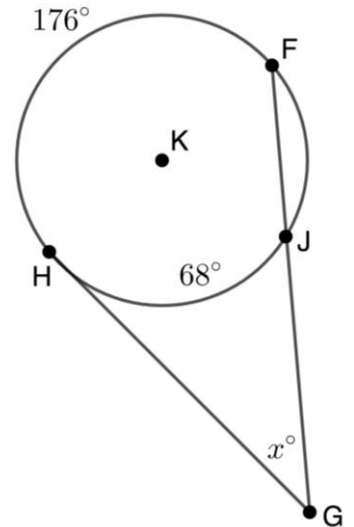
For questions 5-6, Circle V has points T, Q , and R on the circle. Tangent $\overline{TY}$ and secant $\overline{RY}$ intersect at point Y outside the circle. The $m\widehat{TR} = x^\circ$, $m\widehat{TQ} = 72.3^\circ$, and $m\angle TYQ = 53.6^\circ$.



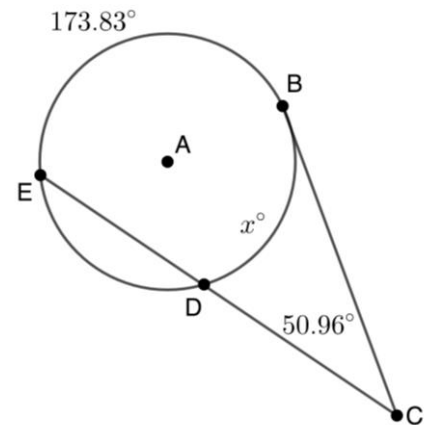
- Write an equation that can be used to find the value of x .

- Find the value of x .

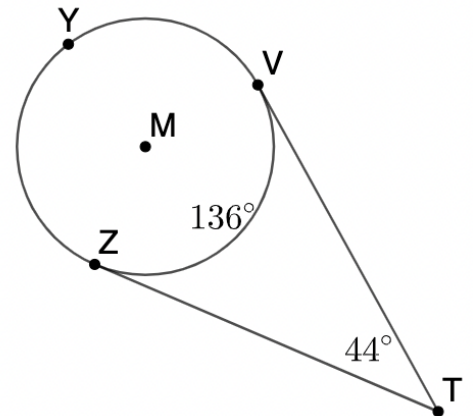
7. Circle K has points F, H , and J on the circle. Tangent $\overline{HG}$ and secant $\overline{FG}$ intersect at point G outside the circle. The $m\widehat{HF} = 176^\circ$, $m\widehat{HJ} = 68^\circ$, and $m\angle JGH = x^\circ$. What is the $m\angle JGH$?



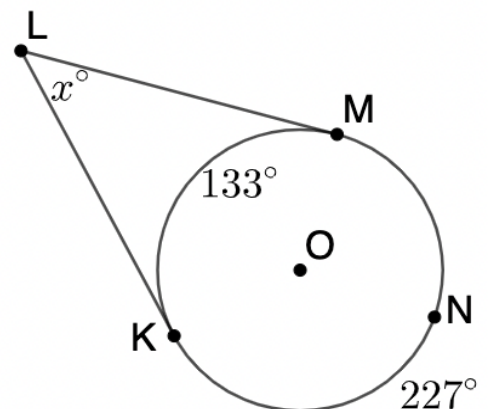
8. Circle A has points B, D , and E on the circle. Tangent $\overline{BC}$ and secant $\overline{EC}$ intersect at point C outside the circle. The $m\widehat{EB} = 173.83^\circ$, $m\widehat{DB} = x^\circ$, and $m\angle BCD = 50.96^\circ$. What is the value of x ?



9. Circle M has points V, Y , and Z on the circle. Tangents $\overline{ZT}$ and $\overline{VT}$ intersect at point T outside the circle. $m\widehat{VZ} = 136^\circ$, and $m\angle VTZ = 44^\circ$. What is the $m\widehat{VYZ}$?



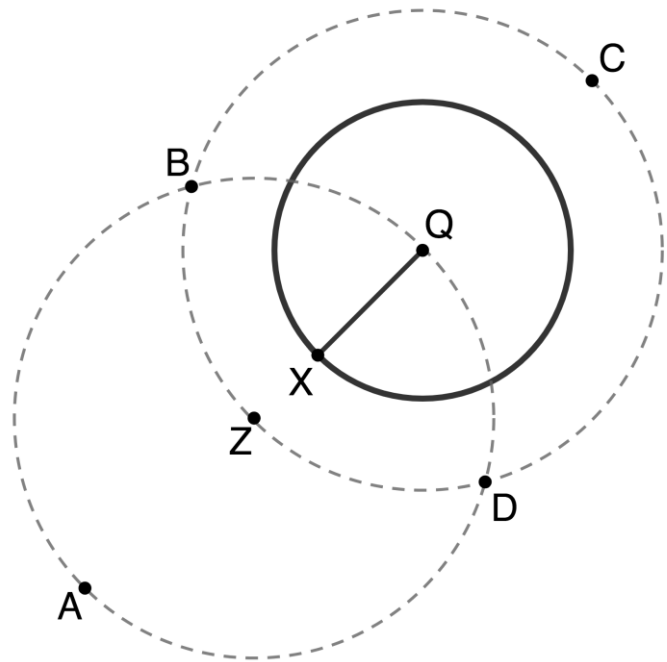
10. Circle O has points K, M , and N on the circle. Tangents $\overline{KL}$ and $\overline{ML}$ intersect at point L outside the circle. The $m\widehat{MNK} = 227^\circ$, $m\widehat{MK} = 133^\circ$, and $m\angle KLM = x^\circ$. What is the $m\angle KLM$?



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Circle Q is shown with radius $\overline{QX}$. The two dotted circles were constructed using a compass so that they are congruent to each other. Points $C, B, D,$ and Z are on the dotted circle Q and points $B, Q, D,$ and A on the dotted circle Z .



1. Select all of the line segments that are congruent to each other.
 - ☐ $\overline{BC}$
 - ☐ $\overline{BZ}$
 - ☐ $\overline{BQ}$
 - ☐ $\overline{QD}$
 - ☐ $\overline{QC}$
 - ☐ $\overline{QZ}$
 - ☐ $\overline{QX}$

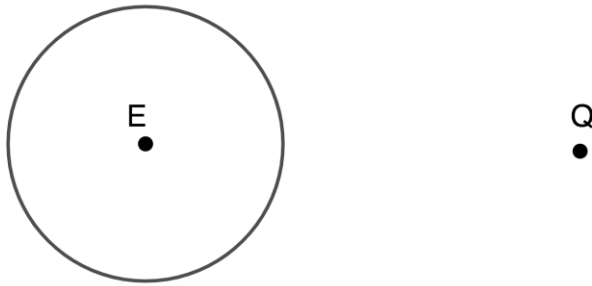
2. Justify why the line segments chosen in question 1 are congruent.

3. Select all of the line segments that are congruent to each other.
 - ☐ $\overline{BD}$
 - ☐ $\overline{AB}$
 - ☐ $\overline{BZ}$
 - ☐ $\overline{AD}$
 - ☐ $\overline{AZ}$
 - ☐ $\overline{ZD}$
 - ☐ $\overline{ZQ}$
 - ☐ $\overline{ZX}$

4. Justify why the line segments chosen in question 3 are congruent.

5. Draw $\overleftrightarrow{AC}$ on the diagram.
6. Draw $\overleftrightarrow{BD}$ on the diagram.
7. Describe how $\overleftrightarrow{AC}$ and $\overleftrightarrow{BD}$ are related.
8. Justify whether $\overleftrightarrow{BD}$ is tangent to circle Q .

Circle E and point Q are shown.



9. Construct the perpendicular bisector of $\overline{EQ}$.
10. Construct a line that is tangent to Circle E and goes through point Q .